

chapter 7

DEMAND FOR HEALTH CAPITAL

Tut-1
40

Explain the demand for health and health care according to Grossman and comparison between traditional demand and health care demand.

Grossman used the theory of human capital to explain the demand for health and health care. According to human capital theory, individuals invest in themselves through education, training and health to increase their earnings. Grossman shows the way in which many important aspects of health demand differ from the traditional approach to demand:

1. It is not medical care as such that the consumer wants, but rather health. People want health, they demand medical care inputs to produce it.

2. The consumer does not merely purchase health passively from the market. Instead, the consumer produces health, combining time devoted to health improving efforts with purchased medical inputs.

3. Health lasts for more than one period. It does not depreciate instantly and can be analyzed like a capital good.

4. Perhaps, most importantly, health can be treated both as a consumption good and an investment good. As a consumption good, health is desired because it makes

people feel better. As an investment good, health is desired because it increases the number of healthy days available to work and to earn income.

Explain the concepts of health capital

Following figure provides a simple schematic diagram that explains the concept of health capital. Consumers apply sets of health inputs which might include not only market inputs of health care but also diet, exercise and time to their physical make-up thus making investments in health capital. These investments maintain or improve the consumers' stocks of health which in turn provide them ^{with} healthy days. Over time, the health stock may grow, remain constant or decline, either slowly with age or more with illness or injury.

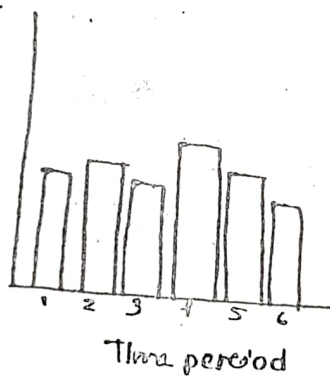
From the figure we see how the end goal, "healthy days" guides consumer decisions as to how much time and money to invest in health stock. We will see that

The prices of health care, the people's wage rates, and their productivities in producing health, will decide how resources are to be allocated among health capital and other goods and services that people buy.

Health inputs

Health care
Diet
Exercise
Environment
Income
Time

Health capital stock
over time



Health outputs
each year

Healthy days
Physical health
Mental health
Activity limitation



Fig. Investing in health capital

Consider a consumer, Edward, who buys market inputs (e.g. health care, food, clothing) and combines them with his own time to produce the stock of health capital that produces services that increase his utility. Edward uses market inputs and personal time both to invest in his

stock of wealth and to produce other things that he
likes.

These other items include virtually all other things
that Edmond does. They include time spent watching
television, reading, playing with and teaching his children,
preparing meals, baking bread, a composite of other
things people do with leisure time. We may call this
composite home good.

subtotal

tut-1
4(b)

43
3/8

Explain the health investment function and home good function in the light of Grossman model.

An increment to capital stock, such as health is called an investment. Consider a consumer Edward, who buys market inputs (e.g. medical care, food, clothing) and combines them with his own time to produce the stock of health capital that produces services that increase his utility. Edward uses market inputs and personal time both to invest in his stock of health ^{and} to produce other things that he likes. During each period, Edward produces an investment in health, I . Thus,

$$\text{Health Investment, } I = I(M, T_H, F)$$

Here, M = market health inputs (providers' service, drugs)

T_H = Time spent improving health.

This function indicates that increased amount of M and T_H increase I .

There are also other items include virtually all other things that Edward does. They include time spent

watching television, reading, playing with and teaching his children, preparing meals, baking bread or watching the sun set, a composite of other things people do with leisure time. we shall call this composite home good B.

Home good function, $B = B(x, T_B)$

Here, x = market purchased goods.

T_B = Time for producing home good.

Home good (B) is produced with time, T_B and market purchased goods, x . This function indicate that increased amounts of x and T_B increase B.

Thus, Edward is using money to buy health care inputs, M or home good inputs, x . He is using leisure time either for health care (T_H) or for producing the home good (T_B).

In this model, Edward's ultimate resource is his own time. Assume that, Edward has 365 days available in the year. To buy market goods, such as medical care, or other goods x , he must trade some of his time for income.

that he must work at a job. Call his time devoted to work T_w , we realize that some of his time during each year will be taken over by ill health or T_L . Thus, we account for his total time in the following manner:

$$\text{Total time} = T = 365 \text{ days} = T_H (\text{Improving health}) + T_B (\text{Producing home goods}) + T_L (\text{lost to illness}) + T_w (\text{working})$$

$$I = I(T, T_H, E)$$

$$B = B(T, T_B, E)$$

$$365 = T_H + T_B + T_w + T_L$$

Question

Test - 1
(50)

How would you explain the potential use of a person's time by the labor-leisure trade off?

The potential uses of a person's time are illustrated by the labor-leisure trade off. In the following figure, the X-axis represents Ed's work and leisure time. Suppose that a person considers his time spent creating health investment to be "health-improvement time" and that he calls T_h his leisure time. Assume further that the number of days lost to ill health and the number of days spent on health enhancing activity are fixed variable. T_{Lo} = Time lost

T_{Ho} = Time spent on healthy activities.

The maximum amount of time that he has available to use either for work, T_w or leisure, T_b 's,

$$\begin{aligned}\text{Time available for work or leisure} &= 365 - T_{Ho} - T_{Lo} \\ &= T_w + T_b\end{aligned}$$

leisure time T_b is measured toward the right while time spent at work, T_w is measured toward the

Let if the person chooses leisure time OA , then he has simultaneously chosen the amount of time of work indicated by AS . These are indicated in the following figure

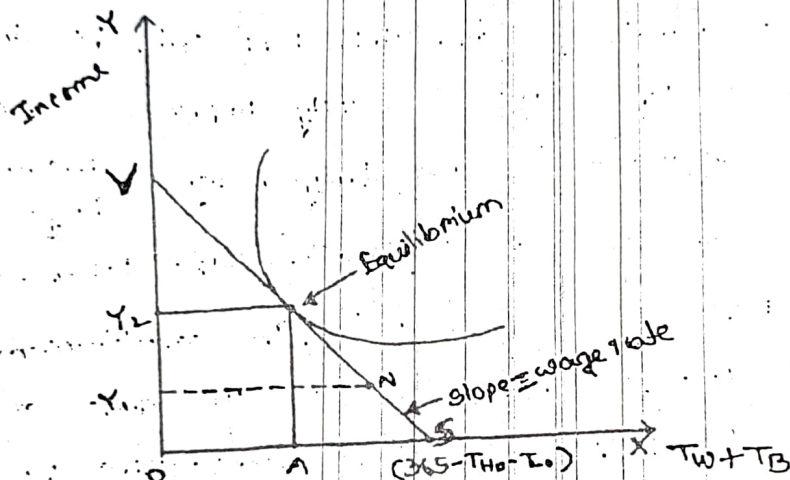


Fig: Labor-leisure trade off

The person's total amount of time available for either work or leisure is given by point S. If he choose point S for the period, he would choose to spend all his available time in leisure. The X-axis represents Income which can be obtained through work. This income can be used to purchase either market health goods or other market goods. Thus if he chooses point S, he will not be able

purchase market, goods because he has no wage.

If the person gives up one day of leisure by spending that day at work, to point A, he will generate income equal to OY_1 , which represents his daily wage. In economic terms, this quantity represents income divided by days worked, that is the daily wage = $\frac{\text{Income}}{\text{Days worked}}$.

The slope of the line VS depicting the labor-leisure trade off reflects the wage rate. In equilibrium, his trade off of leisure and income is the same as the market trade off which is the wage rate. Hence, he takes an amount of leisure and trade AS amount of leisure for income OY_2 .

tutorial

Tut 2
(12)

Explain the impact of an increase in activities to improve health on income and leisure.

An increase in activities to improve health, increased the time spent investing in health status. Time spent on health producing activities, T_H is increased from T_{H0} to T_{H1} . Correspondingly suppose that the number of days lost to ill health has been reduced from T_{L0} to T_{L1} . Time spent on producing health reduces time available for other activities. On the other hand time spent on health investment increases health stock and in turn reduces the time lost to illness.

If the net effect is a gain in available time, then this illustrates the pure investment aspect of health demand. The health investments "pay off" in terms that not only add to potential leisure but that also increase the potential time income, shifting the income-leisure line outward from VS to RQ . The expenditure of time for health producing activities may later improve available

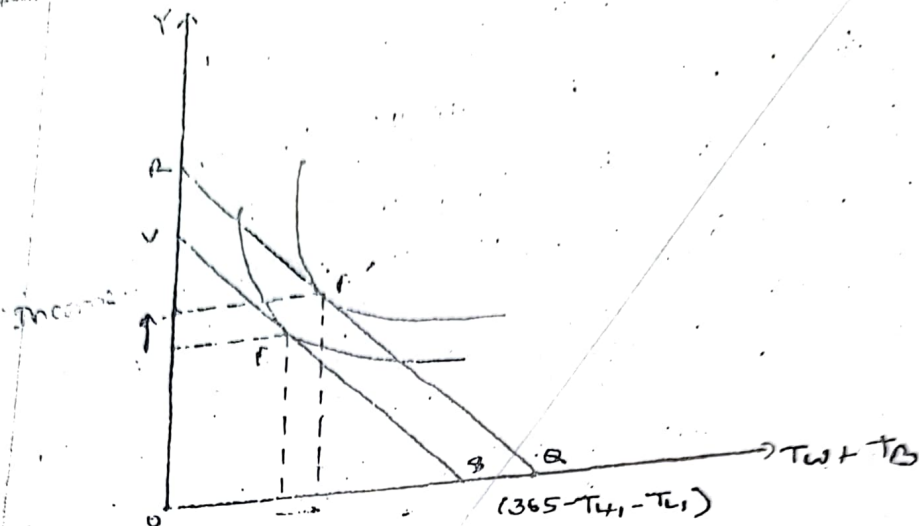


Fig: Increased amount of healthy time due to investment

As a result of investment, utility increases moving from point F to point E. Not only does investment in health lead to feeling better, but also leads to more future income and may lead to more leisure as well.

The improvement in health status also might increase productivity of work, perhaps resulting in a higher wage.

Part-1
4 (C)

Production of healthy days...

Health is a productive good that produces a single output, healthy days. This production function relationship is illustrated in the following figure. The horizontal axis measures health stock in a given period. The greater one's stock of health, the greater the number of healthy days, up to a natural maximum of 365 days. The bowed shape of the curve illustrates the law of diminishing marginal returns. The concept of a health stock minimum shown as H_{min} . At this point, production of healthy days drops to zero indicating death.

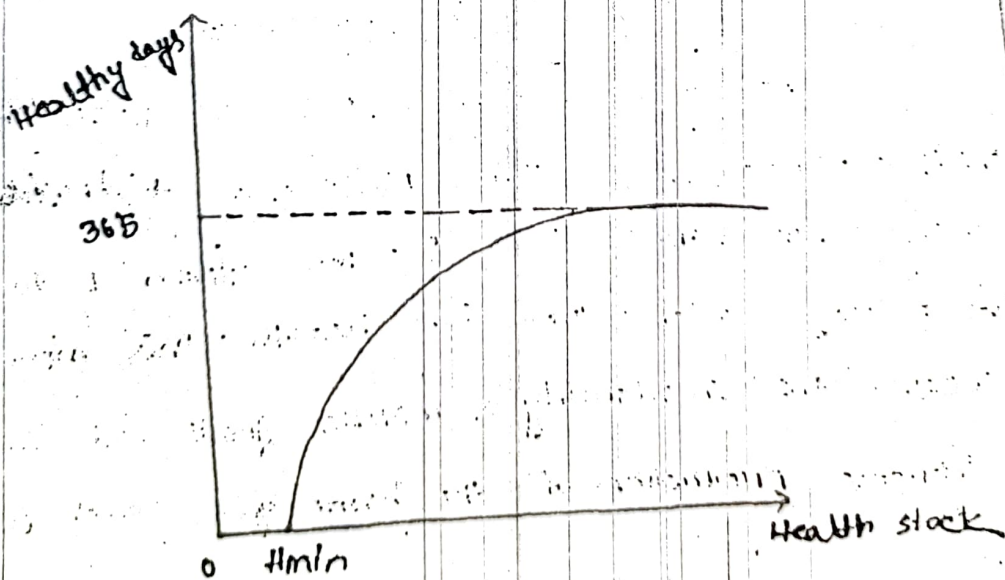


Fig: Production of healthy days.

Subject

Production of health and home goods / Allocation of resources between health and market good (such as Bread)

Consider the possibilities for producing health, H , and home good B given the amount of time variable. Following figure shows the production possibilities trade-off. The curve differs from the usual production possibilities curve in several respects. First, from point A to point C , health improvements increase the amounts of the home good B and health H attainable. It is necessary to increase health beyond H_{min} in order to obtain income and leisure time from which to produce B .

Moving along the production possibilities curve shifts the uses of available time and distributes the purchases of market goods. The move from E to C indicates that more available time for health. This movement also increases the availability of market goods and time used to increase production of the home good such as baking bread.

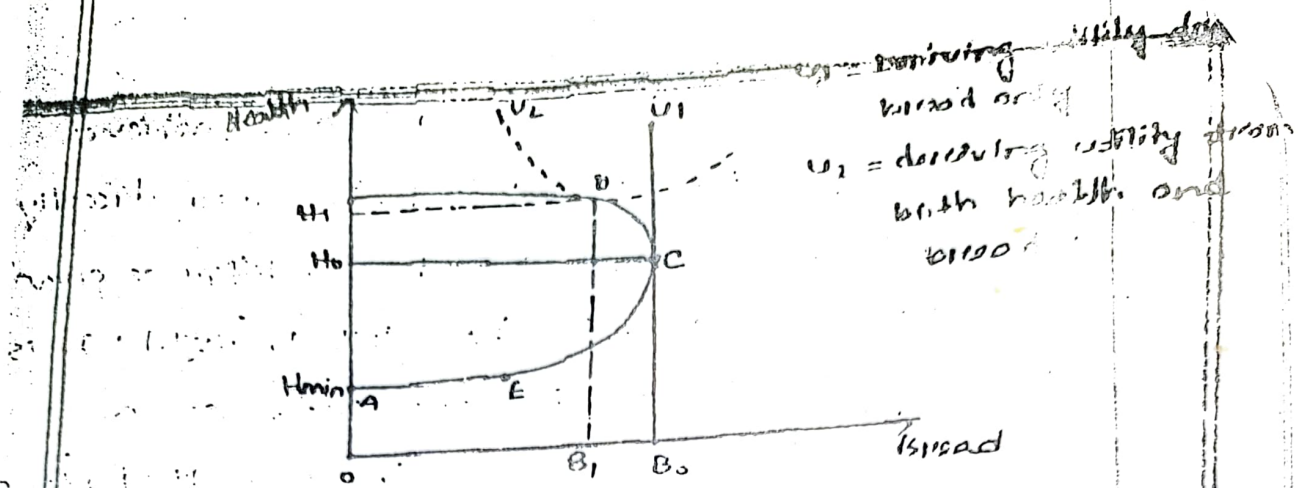


Fig: Allocation of resources between health and bread -

Suppose that a person desires health solely for its effect on the ability to produce income and the leisure time to produce the home good bread. This would imply that his indifference curves between H and B would be vertical lines. In such a case he would maximize his utility by producing as much B as possible. The utility maximizing choice would be at point C , a point of tangency between indifference curve U_1 and the production possibilities curve. He produces amount B_0 of the home good and H_0 of health.

Now assume instead that Ed achieves utility not only from producing B, but also directly from health itself. In this case his indifference curve is sloping downward from left to right. It is more realistic to say that he values health as a consumption good and an investment in productive capacity. The former suggests that he enjoys feeling healthy, the latter that feeling healthy makes him more productive, thus allowing him to earn more.

Continuing...

Chapter 7 Demand for Health Capital

The Cost of Capital:

We demand health because it helps us earn more and feel better. Suppose that an x-ray machine costs \$100,000 and that its price does not change over time. Suppose that the annual income attributable to the ~~use~~^{use} of the x-ray machine is \$20,000. Consider the alternative: Instead of purchasing the x-ray machine, the clinic ~~can~~ could have put the \$100,000 in a savings account, at 5 percent interest rate, yielding the following:

$$\begin{aligned} 100,000 \times 1.05 &= 105,000 \text{ at the end of Year 1} \\ 105,000 \times 1.05 &= 110,250 \text{ at the end of Year 2} \\ 110,250 \times 1.05 &= 115,763 \text{ at the end of Year 3} \\ 115,763 \times 1.05 &= 121,551 \text{ at the end of Year 4} \\ 121,551 \times 1.05 &= 127,629 \text{ at the end of Year 5} \end{aligned}$$

For the investment in an x-ray machine to be desirable & these criteria, it should provide at least \$127,629 in incremental revenue over the five years.

Suppose that the clinic knows that the x-ray

machine will wear out, so that it will be worth only half its original value in five years. The clinic must earn enough not only to cover the opportunity cost from the bank, but also to maintain the value of the machine. For an investment in an x-ray machine to be worth while when it must not only earn the competitive 5 percent return per year but it also must provide enough return to cover depreciation. This suggests that the cost of holding this capital good for any one year as well as over time, will equal the opportunity cost of the capital (interest rate foregone) plus the depreciation.

Marginal efficiency of investment (MEI) and rate of return Optimal health stock determination

The marginal efficiency of investment curve shows how responsive investment is to a change in interest rates. The MEI can be described in terms of the x-ray machine example. A busy clinic may wish to own more than one x-ray machine. The first machine purchased will yield a return of \$20,000. The rate of return is $\frac{\$20,000}{\$100,000} = 20$ percent per year. The management would buy this machine if the incremental revenue brought in covered its opportunity cost of capital and the depreciation. In terms of rates, management would choose to own the first x-ray machine as long as the rate of return, 20 percent, were greater than the interest rate plus the depreciation rate.

If the management considered owning two machines, it would discover that the rate of return on second x-ray machine would probably be less than the first. This is

best understood by recognizing that a clinic buying only one x-ray machine would assign it to the highest priority uses, those with the highest rate of return.

If the clinic were to add a second x-ray machine, then logically it could be assigned only to lesser priority uses. Thus, the second machine would have a lower rate of return than the first. The clinic would then purchase the second machine only if its rate of return were still higher than interest plus depreciation.

The decreasing MEI:

In the following figure, the MEI curve describes the pattern of rates of return, declining as the amount of investment increases. The cost of capital (interest rate, r + the depreciation rate, δ) is shown as the lower horizontal line, labeled $(r + \delta)$. The optimum amount of capital demanded is thus H_0 , which represents the amount of capital at which the marginal efficiency of investment just equals the cost of

capital: This equilibrium occurs at point A.

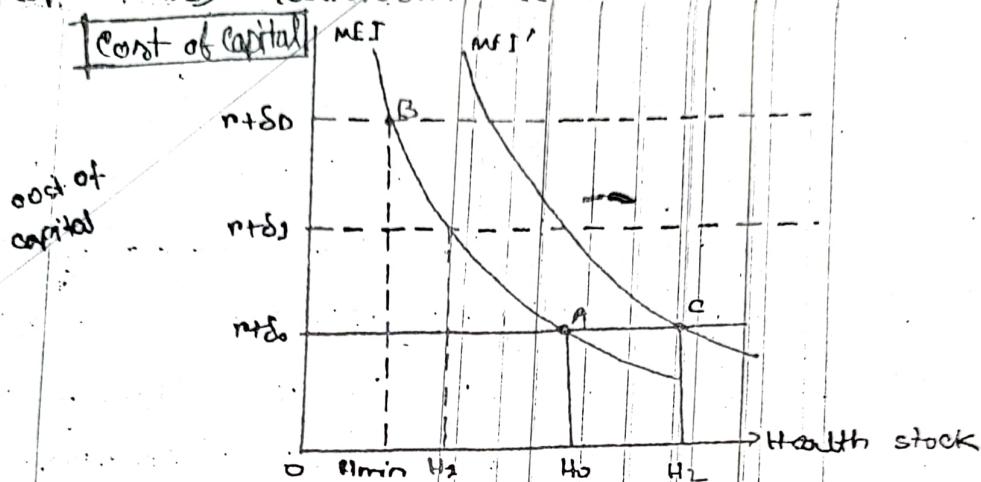


Fig: Optimal health stock.

The MEI curve for investments in health also would downward sloping. This occurs because the production function for healthy days exhibits diminishing marginal returns. The cost of capital for health would similarly reflect the interest rate plus the rate of depreciation health. Understand that a person's health like any capital good, also will depreciate over time. Thus, the optimal demand for health is likewise given at the intersection of the MEI curve and the cost of capital curve ($r + \delta$).

tutorial

Final - 17 (36) Q2/210

2) Changes in equilibrium: Age, Wage, education and uncertainty.

Age: Age is a key factor for changing health condition. Grossman indicates Health stock may depreciate faster during some periods of life and more slowly during others. Eventually as human ages, the depreciation rate is likely to increase. In other words, the health of older people is likely to deteriorate faster than the health of younger people.

Consider the effect of aging on optimal health stock: By hypothesis the depreciation rate δ increases with age from δ_0 to δ_1 and ultimately to δ_2 . These imply that the optimal health stock decreases with age. Optimal health at the younger age, H_0 is greater than H_1 , the optimal stock at the older age. Higher depreciation rates increases the cost of holding health capital stock. In old age, health depreciation rates are extremely high, δ_2 and optimal health stock falls to H_{min} at point B.

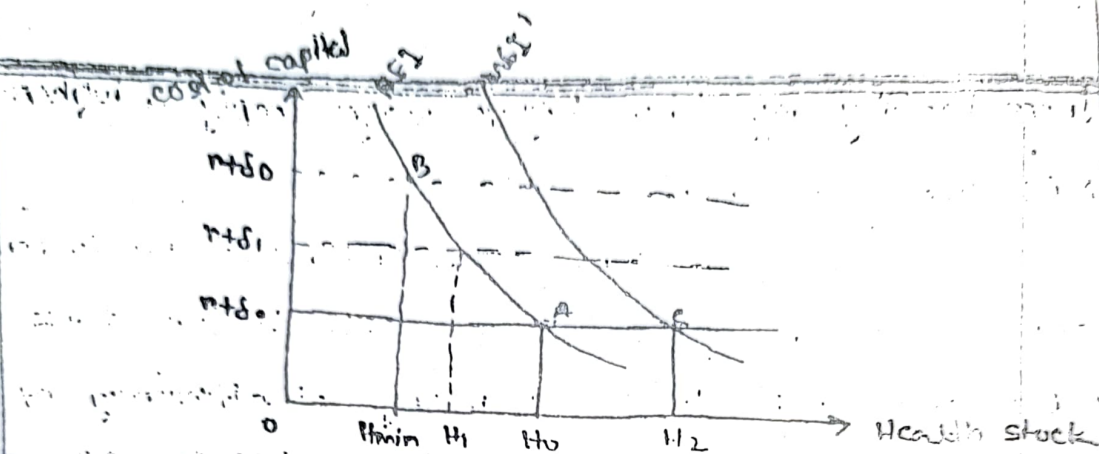


Fig: Optimal health stock

As the expected length of life decreases, the M-EI curve shifts to the left, because the returns from an investment will last for a shorter period of time. Other features of the model suggest that people will increase their gross investment in health as they age. This suggests, in turn, that the elderly would demand more medical care than the young.

Wage rate:

The above figure illustrates the effect of a change in the wage rate on Optimal level of investment. Increased wage rates increases the returns obtained

from healthy days. Thus higher wages imply a higher MEI curve or MEI' .

The original MEI curve describes the lowest wage rate and yields optimal stock health stock, H_0 . MEI' above MEI shows the marginal efficiency of investment for someone with higher wages. At new equilibrium point C , the higher wage will imply a higher optimal level of health stock, H_2 . When person receives this wage effectively drops to zero.

Education

Education is of special interest in the demand for health. Higher education is most often related to better health and most economists believe that education may improve the efficiency with which one can produce investments to health and the home good.

The effect of education is illustrated in the figure. Here the MEI curve illustrates the marginal efficiency of

investment for the consumer with a low level of education while the 'MEI' curve illustrates the same person with a higher level of education. This model indicates that because education raises the marginal product of the direct inputs, it reduces the quantity of these inputs required to produce a given amount of gross investment.

It follows that given investments in health can be generated at less cost for educated people and that they experience higher rates of return to a given stock of health. The health, as shown is that the more educated people will choose higher optimal health stocks, H_2 than the less educated who will choose H_0 . Educated people tend to be significantly healthier.

Educated people are also likely to recognize the benefits of improved health. They may enjoy eating nutritious food or doing physical exercise. They may recognize the dangers of smoking, and the long term